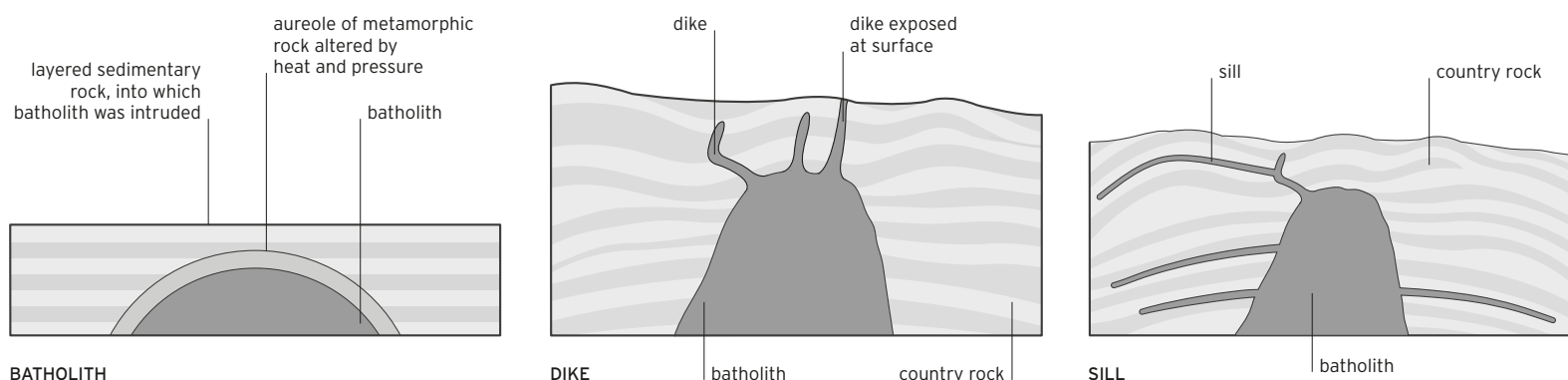


IGNEOUS INTRUSION

An igneous intrusion is a mass of magma that forces its way into Earth's crust and cools and solidifies before reaching the surface. There are three main types: batholiths, dikes, and sills. Batholiths are large igneous bodies that may force the crust up into a dome. Dikes and sills are tabular sheets of igneous rock that have been intruded between layers of sedimentary, metamorphic, or older volcanic

rocks. Dikes are vertical or have a high angle of elevation, whereas sills are horizontal or have a low angle of elevation. Granite, gabbro, diorite, and pegmatite are common intrusive igneous rocks. Over time, the erosion of softer surrounding rocks (referred to as country rock) may expose an intrusion at the surface.



IGNEOUS ROCK Rock formed from the solidification of molten magma.

INFRARED Invisible radiation with a long wavelength, which we feel as heat. See also *ultraviolet*.

INSELBERG An isolated, steep-sided hill or small mountain rising up out of a flat plain. The Sugarloaf Mountain in Rio de Janeiro is an inselberg.

INTRUSIVE IGNEOUS ROCK A rock that formed due to the slow cooling of magma beneath Earth's surface. Examples include diorite and granite. See also *granite*, *igneous intrusion* (panel).

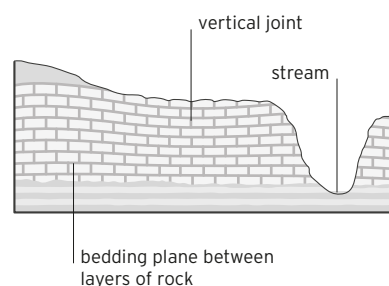
ISLAND ARC A chain of volcanic islands close to a subduction zone. Typically, there is a deep ocean trench on one side. See also *back arc*, *subduction*, *volcanic arc*.

ISOBAR An imaginary line connecting points with the same air pressure.

ISOSTASY The concept that continents rise high above the sea floor because continental crust is less dense than oceanic crust.

J

JET STREAM A ribbon of very strong, high-altitude winds, which move weather systems around Earth.



JOINT ▲ A fracture dividing rock into sections. The rocks on either side of a joint have not moved relative to each other, unlike the rocks either side of a fault. See also *fault* (panel).

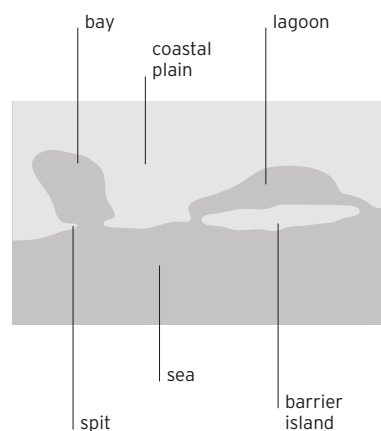
K

KARST A landscape, usually found in limestone regions, characterized by caves, sinkholes, and underground watercourses. See also *limestone*, *sinkhole*.

KETTLE LAKE A water body occupying a kettle hole, a depression left after an ice block from a former glacier has melted.

L

LA NIÑA The phenomenon in which the waters of the eastern Pacific Ocean become unusually cold. This is the opposite of the El Niño effect. See also *El Niño*.



LAGOON ▲ An expanse of seawater that is almost cut off from the open ocean and consequently relatively sheltered. See also *atoll*.

LAHAR A flow of mud, water, ash, and other debris down the slopes of a volcano. See also *mass movement*.

LAMINATION Fine and generally parallel layers within rocks. It is usually less distinct than bedding. See also *bedding*.

LATENT HEAT The heat that is released, for example, when water vapor condenses into water droplets. This heat release is an important factor in the growth of storms. See also *condensation*.

LATERAL MORaine The ridges of silt, sand, and gravel that build up along the sides of a glacier. They indicate the extent of ice before the glacier retreated. See also *moraine* (panel).

LAVA Molten rock that has reached Earth's surface as a result of an eruption. See also *basalt*, *pillow lava*.

LEACHING The process of rainwater washing minerals and nutrients through topsoil. Leached materials tend to be lost from the topsoil and deposited in the subsoil.